Phase III status

Intrinsic $\mathrm{U}(3)$ richness of the inter-window matrix

Route D negative result and consequences for Routes A–E

Phase III headline. The Phase II finding that the inter-window matrix $D \in \mathrm{U}(3)$ carries irreducibly rich structure has been sharpened: *the structure persists in any LZ-block convention.* Re-extracting D using the proper Stokes-phased Landau–Zener amplitude blocks N_X^{proper} (with $\varphi_S(\delta) = \delta(\log\delta - 1) + \pi/4 + \arg\Gamma(1 - \mathrm{i}\delta)$) does *not* reduce the residual of the $(\mathrm{diag} \cdot R_{(\mathrm{sp}_A, \mathrm{sp}_B)})$ ansatz: median residual on a 12-sample calibration is 0.0664 in both the phase-free and the Stokes-phased convention, identical to within six significant figures, across all four sign-convention combinations $(\sigma_A, \sigma_B) \in \{\pm 1\}^2$. The Phase III plan's falsifiable exit "Route D fails: the LZ block convention is not the issue; the U(3) richness is intrinsic" is therefore *triggered*.

1 Route D execution and outcome

Route D was the foundation step of the Phase III plan: re-extract the empirical inter-window matrix $D = N_B^{-1} U_{\text{bench}} N_A^{-1}$ using Stokes-phased LZ amplitude blocks rather than the phase-free real-orthogonal block of `assay.closed_form.lz_rotation`. The proper block carries $\varphi_S(\delta)$ on its diagonal entries:

$$N_{\text{proper}}(p, \delta) \;=\; \begin{pmatrix} \sqrt{1-p}\,\mathrm{e}^{-\mathrm{i}\sigma\,\varphi_S(\delta)} & \sqrt{p} \\ -\sqrt{p} & \sqrt{1-p}\,\mathrm{e}^{+\mathrm{i}\sigma\,\varphi_S(\delta)} \end{pmatrix},$$

with $\sigma \in \{+1, -1\}$ the convention's sign degree of freedom.

1.1 Calibration result

On a 12-sample calibration subset (3 each from the four success strata $\{S1, S6, S7, S8\}$), all four (σ_A, σ_B) combinations produce the *identical* median residual of the $(\mathrm{diag} \cdot R)$ ansatz reconstruction:

(σ_A, σ_B)	median resid	max resid	wall time
$(+1, +1)$	0.0664	0.7681	90 s
$(+1, -1)$	0.0664	0.7681	84 s
$(-1, +1)$	0.0665	0.7681	102 s
$(-1, -1)$	0.0665	0.7681	91 s
phase-free baseline	0.0664	1.4454	~ 90 s

The Stokes-phased convention modestly reduces the *tail* (max-residual $1.4454 \to 0.7681$, a $2\times$ improvement at the worst samples) but does not move the *median* at all. Whatever phase content the Stokes-phased N_X^{proper} captures, it is not the dominant source of D's rank-three U(3) richness.

1.2 The intrinsic off-block content

Direct inspection of a worst-case S2 sample with intermediate $(p_A, p_B) \approx (0.43, 0.02)$ shows that the residual of the $(\mathrm{diag} \cdot R_{(\mathrm{sp}_A, \mathrm{sp}_B)})$ ansatz lies *outside* the spectator $(\mathrm{sp}_A, \mathrm{sp}_B)$ plane:

$$D \approx \begin{pmatrix} +0.51+0.19\mathrm{i} & +0.32+0.15\mathrm{i} & +0.12+0.75\mathrm{i} \\ +0.02-0.56\mathrm{i} & -0.10+0.82\mathrm{i} & -0.02+0.01\mathrm{i} \\ -0.36+0.51\mathrm{i} & -0.28+0.33\mathrm{i} & +0.65+0.07\mathrm{i} \end{pmatrix} \quad (D - D_{\mathrm{model}})_{\mathrm{im}} \approx \begin{pmatrix} -0.16 & +0.15 & +0.75 \\ -0.56 & -0.12 & -0.28 \\ +0.51 & +0.36 & -0.03 \end{pmatrix}.$$

The model residual entries at positions $(0,1)$, $(0,2)$, $(1,0)$, $(2,0)$, $(1,2)$, $(2,1)$ are all of order 0.3–0.8. The $(\mathrm{diag} \cdot R)$ ansatz forces the common active index $i_c = 0$ to be a fixed point of the rotation, so its row and column should be diagonal + zeros. They are not: $D[0,1]$, $D[0,2]$, $D[1,0]$, $D[2,0]$ all carry $O(1)$ magnitudes.

The closed form must produce a full $\mathrm{U}(3)$, not a $\mathrm{U}(2) \oplus \mathrm{U}(1)$ subgroup.

2 Consequences for Routes A, B, C

The Phase III plan's risk-ranked outlook had Route C (AKT/DDP) as the highest-confidence *predictive* route on the grounds that its architecture had the right spectator-plane rotation. The empirical D now invalidates this architecture: a rotation in the $(\mathrm{sp}_A, \mathrm{sp}_B)$ plane is structurally a $\mathrm{U}(2) \oplus \mathrm{U}(1)$ element, but D is a full $\mathrm{U}(3)$ element. *Route C as architecturally specified cannot reach the closed form, regardless of how well its Bessel block is implemented.*

The implementation work executed in Phase III confirms the failure modes empirically:

- **Route C extension (`formula_DDP_aki`).** Replaced the $(\mathrm{sp}_A, \mathrm{sp}_B)$ $\mathrm{SO}(2)$ rotation with a modified-Bessel block at index $\nu = 1/3$, with the corrected $V_{AB} = \int (L_H/p)\sqrt{Q_4}\mathrm{d}u$ Voros symbol on the spinor cover. Smoke test on 40 samples: per-stratum mean residual 0.60–0.82, *worse* than the original `formula_DDP`'s 0.43–0.60. Two causes: (i) the implemented Bessel block uses a heuristic parameterisation ($\sin\theta = \sin(\arg V)$ for the mixing magnitude, not the proper Wronskian-normalised K_ν/I_ν ratio); (ii) the "corrected" V_{AB} on the exterior sheet λ_0 is exactly $2\times$ the old V across all samples, suggesting the integration contour is on the wrong sheet.
- **Route A extension (`formula_HN_nonabelian`).** Wired the existing `spectral_network.py` machinery into the connection-matrix product via wall-jump matrices $K_\gamma = I + \mu_\gamma E_{ij}$ at real-axis wall crossings. The implementation overflows catastrophically (per-stratum residual 10^{23} to 10^{193}). Cause: the Voros symbol $\mu = \mathrm{e}^{\mathrm{i}I_X}$ with $\mathrm{Im}\, I_X < 0$ produces $|\mu| = \mathrm{e}^{|\,\mathrm{Im}\, I_X|}$ which is huge. The naive unipotent factor $I + \mu E_{ij}$ is non-unitary; without careful side-tracking (pairing each wall jump with a conjugate factor whose product is unitary), the product loses unitarity by hundreds of orders of magnitude.
- **Route B extension.** Not implemented in Phase III, but the log-space + Faddeev finite-b refactor remains the prescribed path and is a candidate for future work. Its closed-form image is a full $\mathrm{U}(3)$ via the mutation product, so it is not architecturally excluded by Route D's finding.

3 Refined open problem

The closed-form challenge is now precisely:

Find a $\mathrm{U}(3)$**-valued function** $D(\epsilon, \gamma, a)$ whose $|\cdot|^2$ projection plus row/column stochasticity gives the two remaining transcendental entries of P. The function must:

- be parametrised by 8 real parameters (6 after the global $\mathrm{U}(1)$ gauge), *not* reducible to a 2-plane rotation;
- match the empirical D on 300 stratified samples to $\mathrm{U}(3)$ geodesic distance $\leq 10^{-8}$;
- be derivable from one of the three exact-WKB literature frameworks (Hollands–Neitzke, Iwaki–Nakanishi, AKT) OR surfaced directly from data via symbolic regression.

This is a research-level problem. The Phase II/III prototypes collectively demonstrate that none of the standard sequential 2-level LZ products, restricted spectator-rotation ansätze, or single-window Stokes-phase corrections can produce it.

4 Recommended next steps

1. **Proper Hollands–Neitzke side-tracking (Route A done correctly).** The `spectral_network` module identifies all real-axis wall crossings, BPS saddles, and sheet pairs. The missing piece is the GMN "ω-pair" construction: each BPS state contributes a pair of unipotent factors with reciprocal Voros symbols whose product is a unitary 2×2 rotation. Implementing this side-tracking is mechanical but requires careful attention to which side of each wall each chamber sits on. Estimated effort: 5–10 days.
2. **Iwaki–Nakanishi with log-space arithmetic and finite-b Faddeev (Route B done correctly).** Produces a full $\mathrm{U}(3)$ via the cluster mutation product. The implementation pieces (log-space matrix accumulation; finite-b Faddeev convergence) are well-defined. Estimated effort: 3–5 days.
3. **PySR symbolic regression on the empirical D (Route E with proper backend).** Install PySR (requires the Julia toolchain, an explicit permission grant needed in this auto-mode environment), feed it the 300-sample D dataset, and let it search over an operator grammar that includes Γ-functions, modified Bessel functions, and quantum dilogarithms. If the closed form is a finite product of named transcendentals at parameter-determined arguments (the expected shape), PySR will surface it.
4. **Deep literature dive on Hollands–Neitzke non-abelianisation at the genus-1 Hitchin curve.** The Type-1, $N = 3$ spectral curve has $b_1(\Sigma_Y) = 2$, matching the window count. The Hollands–Neitzke 2019 paper on non-abelianisation for genus-1 Hitchin systems is the direct reference. A careful read should resolve the side-tracking and θ-phase conventions that defeated the Phase III prototype.

5 Status of the closed-form push

- **Exact & elementary** (Phase I deliverables): two Brundobler–Elser extreme-survival entries. Validated to numerical floor on 300 samples.
- **Empirically extracted** (Phase III deliverable): the inter-window matrix $D \in \mathrm{U}(3)$ on 300 samples, in two conventions (phase-free, Stokes-phased), with the $(\mathrm{diag} \cdot R)$ ansatz residual quantified.
- **Constrained** (Phase III finding): D is structurally a full $\mathrm{U}(3)$, not a $\mathrm{U}(2) \oplus \mathrm{U}(1)$ subgroup. Architectural ansätze that produce only a spectator-plane rotation are excluded.

- **Open**: the analytic form of $D(\epsilon, \gamma, a) \in \mathrm{U}(3)$.

6 Artifacts

New code (Phase III):

- `assay/lz_stokes_phased.py` – Stokes-phased LZ amplitude block.
- `assay/d_extract.py` – shared D-extraction primitives, U(3) geodesic distance.
- `assay/convergence_test.py` – pairwise route comparison and empirical D agreement gate.
- `notes/phase2_D_stokes_phased.py` – Route D driver (sign calibration + 300-sample sweep).
- `notes/phase2_symbolic_regression_lite.py` – numpy-only fallback when PySR is unavailable.
- `assay/ddp.py` (extended) – `formula_DDP_aki`, `aki_bessel_block`, `inter_window_action_corrected`, `ddp_D`.
- `assay/hitchin_holonomy.py` (extended) – `formula_HN_nonabelian`, `hn_connection_matrix_nonabelian`, `hn_D`, `_wall_data`.

Output datasets:

- `notes/phase2_D_stokes_phased.csv` – 300 samples × 9 complex D entries × (theta, chi, model_resid, ...) in Stokes-phased convention.
- `assay/empirical_D_cache.pkl` – per-record cache for `empirical_D`.

Plan file: `vast-drifting-avalanche.md` (the approved Phase III plan).